Intelligent Agents

1. Definition of Intelligent Agent:

An intelligent agent is an entity that perceives its environment through sensors and acts upon that environment using actuators to achieve specific goals.

Example: A robot uses cameras (sensors) to see and wheels (actuators) to move.

2. Perception and Action:

An agent continuously interacts with its environment by receiving inputs (percepts) and producing outputs (actions).

- **Percepts** → Information received from environment
- **Actions** → Response given by the agent

Example: A thermostat senses temperature and turns the heater on/off.

3. Agent Function:

The agent function maps the sequence of percepts to an action. It defines how the agent behaves in every possible situation.

👉 Agent Function: *Percept Sequence* → *Action*

Example: If a vacuum cleaner senses dirt → it performs the action "clean".

4. Agent Program:

An agent program is the implementation of the agent function. It runs on a physical system called the architecture.

👉 **Agent = Architecture + Program**

Example:

- Architecture → Robot hardware
- Program → Code controlling robot behavior

5. Types of Agents:

a) Simple Reflex Agents

Act only based on current percept using condition-action rules.

Example: If room is dirty → clean

b) Model-Based Reflex Agents

Maintain an internal state to handle partially observable environments.

Example: Robot remembers where obstacles are.

c) Goal-Based Agents

Take actions to achieve specific goals by considering future outcomes.

Example: GPS finding shortest path to destination.

d) Utility-Based Agents

Choose actions based on a utility function (best outcome among options).

Example: Self-driving car choosing safest and fastest route.

6. Rationality of Agents:

An agent is said to be rational if it selects actions that maximize its performance measure based on its knowledge and percepts.

Example: A vacuum agent cleans efficiently with minimum moves.

7. Performance Measure:

It evaluates how well an agent performs its task.

Example:

- Vacuum cleaner → cleanliness of room
- Self-driving car → safety and efficiency

Key Point

An intelligent agent is a system that perceives its environment, processes information, and takes actions to achieve goals efficiently.